Aaditya Lad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.35
Vidhya Bhavan Public School,Indore	Class XII (CBSE)	2022	93.80%
Sri Sathya Sai Vidya Vihar, Indore	Class X (CBSE)	2020	93.80%

WORK EXPERIENCE

ServiceNow -- Machine Learning Intern | Hyderabad

- Built and deployed multiple **LLM-powered AI agents** for conversational configuration of Now Assist, contributing to a production solution that reached **General Availability (GA) in May 2026**.
- Developed agent workflows capable of translating **natural-language user requests into configuration actions**, automating selection of LLM models, model versions, and AI guardrails.
- Performed iterative testing and evaluation of agent behaviour, improving **intent understanding, action selection, reliability, and error handling** across configuration scenarios.
- Worked with cross-functional engineering teams to take AI-agent capabilities from **development and testing to production deployment**.

PROJECTS

- **Agentic RAG Research Assistant using LangChain, Qdrant and Groq (Llama 3)**

1. Architected a Retrieval-Augmented Generation (RAG) pipeline using Large Language Models (LLMs), combining hybrid retrieval (dense vector embeddings + sparse BM25 keyword search, fused via Qdrant's Reciprocal Rank Fusion) with cross-encoder reranking (FlashRank) and a Corrective RAG (CRAG) agentic loop for automated relevance grading and query rewriting.
2. Built a semantic chunking and contextual retrieval pipeline for unstructured PDF document processing, and developed a systematic evaluation framework using RAGAS to benchmark retrieval quality across faithfulness, answer relevancy, and context precision metrics.
3. Engineered hybrid search and reranking to improve retrieval relevancy and reduce hallucinated responses over a dense-only vector search baseline, validated through reproducible before/after benchmarking.

- **Credit Card Fraud Detection using Machine Learning**

1. Built and compared Logistic Regression, SVM, and Decision Tree models on 284,807 transactions (0.17% fraud cases), using undersampling and class weighting to handle extreme class imbalance.
2. **Decision Tree:** 77% recall, 70% precision, 99.9% overall accuracy, detecting 103/134 frauds in test set.
3. **SVM:** 86% recall, 98.9% precision, F1-score 92.1%, 94.6% training accuracy, balancing high detection and minimal false positives.
4. **Logistic Regression:** 91.4% test accuracy, 94.4% training accuracy, demonstrating solid baseline performance.
5. Optimized all models via cross-validation, ROC/F1 analysis, and hyperparameter tuning, prioritizing recall to minimize undetected frauds.

RELEVANT COURSES

- **Introduction to Programming (ID1063)**

Studied programming fundamentals, algorithmic logic, and hands-on problem-solving using the C language.

- **Machine Learning Specialization - Deeplearning.ai *(by Andrew Ng)***

Completed a 3-course series: Supervised Machine Learning, Advanced Learning Algorithms, and Unsupervised Learning.

SKILLS

- **Programming & Data Handling:** Python, C++, C, SQL, NumPy, Pandas, Matplotlib, Seaborn
- **Generative AI & LLMs:** LangChain, Retrieval-Augmented Generation (RAG), Large LanguageModels (LLMs), Vector Databases (Qdrant), Hybrid Search, Prompt Engineering, LLMEvaluation (RAGAS), FastAPI, Streamlit
- **Machine Learning & AI**: scikit-learn, TensorFlow, Keras, XGBoost, Convolutional NeuralNetworks (CNNs), Logistic Regression, SVM, Decision Trees
- **Advanced Topics:** Time-Series Forecasting, Ensemble Models, Feature Engineering,Cross-Validation, Hyperparameter Tuning
- **Data Science Tools & Platforms:** Jupyter Notebook, Google Colab, VS Code, Anaconda
- **Software Engineering & MLOps:** Git, GitHub, Model Evaluation (ROC, F1, AUC), A/B Testing,Model Monitoring & Documentation
- **Computer Science Foundations:** Data Structures & Algorithms, Operating Systems,Computer Networks, Database Management Systems (DBMS)
- **Soft Skills:** Problem-Solving, Communication, Adaptability

POSITIONS OF RESPONSIBILITY

- **Corporate Relations & Marketing Coordinator, Tinkerer's Lab**

Led corporate outreach and marketing initiatives to secure industry collaborations and sponsorships for innovation projects and the flagship event.

- **Hospitality Coordinator, Elan & Vision**

Managed logistics of guests, artists, and dignitaries during the institute's flagship techno-cultural fest.